

EXPERIMENT 5

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AIM-Simulation of Sound Level Detector using Tinker cad.

THEORY-

Tinkercad is an online simulation tool that allows users to design, simulate, and test electronic circuits. It provides a virtual environment where components like Arduino, sensors, LCDs, LEDs, motors, and more can be connected and programmed without needing physical hardware.

Tinkercad is widely used for:

- Learning and testing Arduino projects.
- Simulating sensor-based applications.
- Debugging circuits before implementing them in real life.

Since Tinkercad does not have built-in electret microphone sensor so we will be using piezo element, its acts as a simple microphone by sound waves into electrical waves. When sound waves hit the piezo element, it vibrates, generating a small electrical voltage proportional to the intensity of the sound.

Objectives of Simulation

- Verify functionality of the electret microphone.
- Test LED responses to different sound levels.
- Validate conditional logic for LED control.
- Debug and monitor using the Serial Monitor.
- Optimize sound level thresholds.
- Evaluate system stability.

Steps-

● Step 1: Open Tinkercad and Create a New Circuit

1. Go to Tinkercad (<https://www.tinkercad.com/>).
2. Click on "Circuits" from the left panel.
3. Click "Create New Circuit".

● Step 2: Add Components

Search and drag the following components onto the workspace:

1. Arduino Uno
2. Piezo Element (as a sound sensor)
3. LEDs (Red, Yellow, Green)
4. Resistors (220 Ω for LEDs)
5. Jumper Wires

● Step 3: Wire the Component

Wiring the Piezo Element to Arduino

Piezo Pin	Connect To
GND	Arduino GND
Signal	Arduino A0

Wiring the LEDs to Arduino

LED Color	Anode (Longer Leg)	Cathode (Shorter Leg)
Red	Arduino Pin 2	220 Ω Resistor to GND
Yellow	Arduino Pin 3	220 Ω Resistor to GND
Green	Arduino Pin 4	220 Ω Resistor to GND

Wiring the Resistors

Ensure each LED cathode is connected to a 220Ω resistor, which is then connected to the ground (GND) rail on the breadboard.

• Step 4: Write the Arduino code

1. Click on Code and select Text mode

```
const int redLED = 2;
const int yellowLED = 3;
const int greenLED = 4;
const int micPin = A0;
void setup() {
  pinMode(redLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(greenLED, OUTPUT);
  pinMode(micPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  int soundValue = analogRead(micPin);
  Serial.println(soundValue);

  if (soundValue > 800) {
    digitalWrite(redLED, HIGH);
    digitalWrite(yellowLED, LOW);
    digitalWrite(greenLED, LOW);
```

```

} else if (soundValue > 500) {
  digitalWrite(redLED, LOW);
  digitalWrite(yellowLED, HIGH);
  digitalWrite(greenLED, LOW);
} else {
  digitalWrite(redLED, LOW);
  digitalWrite(yellowLED, LOW);
  digitalWrite(greenLED, HIGH);
}
delay(100);
}

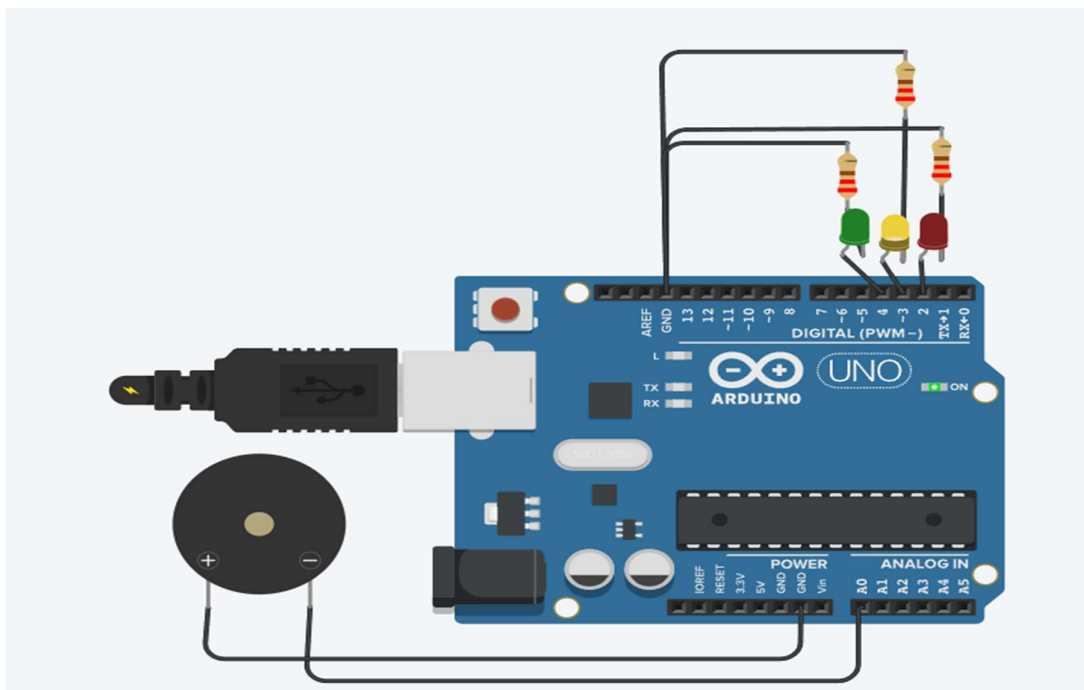
```

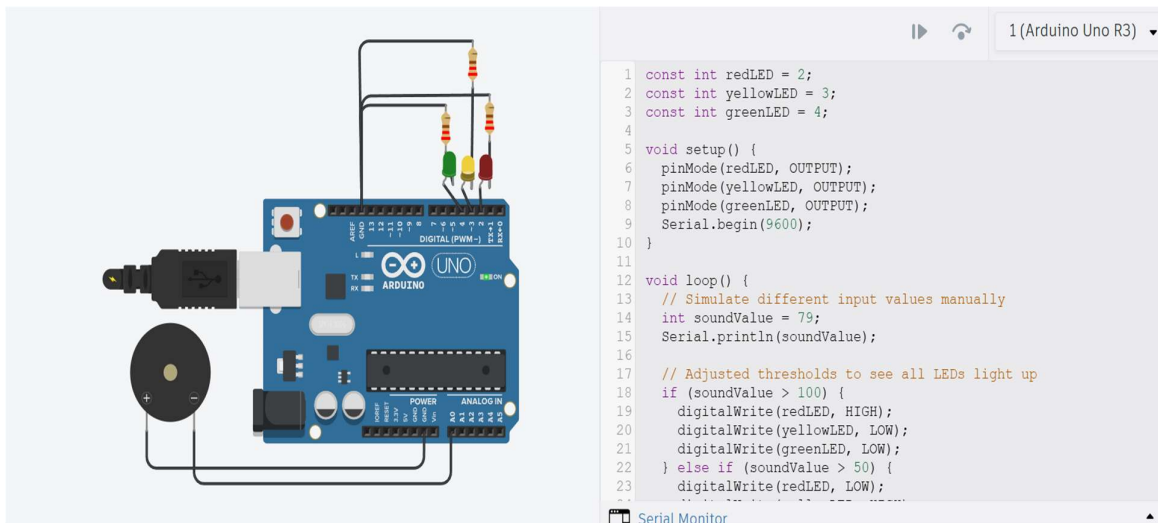
Step 5: Start the Simulation

Power up the Arduino.

Open the Serial Monitor in the Arduino IDE to observe sound values.

Tap or make different levels of noise near the piezo element and observe which LEDs light up in response to the sound intensity.





Conclusion

The simulation successfully demonstrated sound level detection using a piezo element in Tinkercad. The system effectively responded to varying sound intensities by illuminating different LEDs, showcasing how sound sensors can be utilized for interactive displays, noise monitoring, and environmental sensing applications.